Juye Xiao

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Professional Summary

- Data Analyst with nearly 4 years of experience, holding a Master of Science in Applied Analytics from Columbia University and dual bachelor's degrees from the University of Michigan.
- Proven ability to optimize business operations, achieving up to **a 12% reduction in costs** and a **15% increase in marketing ROI** through data-driven decision making and A/B testing methodologies.
- Demonstrated technical leadership and cross-functional communication as the **Tech Lead** for a high-impact, **S&P Global-sponsored** capstone project, enhancing climate risk assessment models.

Core Technical Skills

- **Programming Languages:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), SQL (PostgreSQL, Complex Queries, Window Functions), R
- **Data Analysis & Modeling:** Exploratory Data Analysis (EDA), Statistical Modeling (Regression, Hypothesis Testing, A/B Testing), Time-Series Analysis (ARIMA), Machine Learning (Logistic Regression, Decision Trees, Clustering, Random Forest, XGBoost), NLP (TF-IDF, LDA)
- **Tools & Platforms:** Apache Spark (PySpark), Tableau, Power BI, Git, Jupyter Notebook, Databricks
- **Core Competencies:** Data Visualization, Data Storytelling, Cross-functional Communication

Professional Experience

Code Auto Tools | Los Angeles, US

Data Analyst | Sept. 2023 - Present

- **Inventory & Sales Analysis**
  - Engineered and maintained ETL pipelines using SQL to integrate and process multidimensional sales and inventory data, enhancing data reliability for analysis.
  - Developed ARIMA-based predictive models for key component sales, leading to optimized inventory strategies and a **10-12% reduction** in overstocking costs for high-value product lines.
  - Automated data validation between sales orders and inventory records, providing data-driven inventory alerts and replenishment recommendations to the procurement department.
- **E-commerce & Advertising Analysis**
  - Designed and implemented A/B testing frameworks on eBay and Amazon, applying hypothesis testing to optimize ad copy and target audience segmentation, which **boosted advertising ROI by 15%** for key campaigns.
  - Analyzed user behavior data and customer feedback from multiple e-commerce platforms to identify product design and logistics issues, providing insights that informed operational improvements.
- **Market & Logistics Analysis**
  - Monitored and analyzed competitors' pricing and promotional strategies, delivering competitive analysis reports that guided dynamic pricing adjustments.
  - Analyzed logistics data to identify cost-saving opportunities, proposing alternative solutions that achieved a **5-8% reduction in transportation costs** on specific routes.

China United Transportation Corp | Los Angeles, US

Database Administrator and Operations Analyst | Feb. 2021 - Aug. 2023

Role performed part-time (Sep 2021 – Feb 2023) during Master's studies and full-time otherwise.

- Optimized container allocation and delivery routes by analyzing historical transportation data with SQL and Python, successfully reducing idle waiting times by approximately 8%.
- Led the digital transformation of paper-based order records into a standardized electronic database, **slashing average query time by 70%**.
- Maintained and optimized the transportation business database, which improved the response time of key business queries by **25%**.

DiDi Global Inc. | Beijing, China

Front-End Development Intern | Jun. 2019 - Aug. 2019

- Developed an automated website performance monitoring platform using React and Node.js, which reduced the time to detect performance issues by **40%**.
- Analyzed platform data (e.g., FCP, TTI) to identify performance bottlenecks and proposed optimizations that accelerated average page load speed by **25%**.

Projects

Optimization of Climate Change Monitoring and Prediction System | S&P Global Sponsored

Tech Lead | Sept. 2022 - Jan. 2023

Tech Stack: Python (Data Processing, Statistical Modeling), Linear Regression, Time Series Decomposition, MODIS & NEX-GDDP

- Led the technical analysis of satellite (**MODIS**) and climate model (**NEX-GDDP**) data to enhance **S&P Global**'s climate risk assessment accuracy.
- Utilized linear regression and time series decomposition in Python to quantify a systematic mean temperature deviation of **13.2°C** between the two data sources.
- Discovered that prediction deviations increased under high-emission scenarios, providing critical data for adjusting risk assessment models.
- **Technical Value**: Enhanced reliability of climate data analysis by using model forecasts to reconstruct missing historical satellite data, improving data coverage.

Bank Customer Churn Prediction Model

Individual Project | May 2023 - Oct. 2023

Tech Stack: Python, Scikit-learn (Logistic Regression, KNN, RandomForestClassifier), Pandas, GridSearchCV

- Developed and compared three machine learning models to predict customer churn, processing a dataset of 10,000 customer records.
- Engineered features using One-Hot and Ordinal Encoding and optimized model hyperparameters with GridSearchCV and 10-fold cross-validation.
- Selected Random Forest as the optimal model, achieving **86.1% accuracy and 0.845 AUC** on the test set.
- Identified key churn drivers (Age, Balance, NumOfProducts, etc.) to provide data-driven support for customer retention strategies.

YouTube Comment User Profile Analysis and KOL Identification

Individual Project | Apr. 2024 - Sept. 2024

Tech Stack: Apache Spark, PySpark MLlib, Logistic Regression, TF-IDF, LDA

- Built a data pipeline using Apache Spark and PySpark MLlib to process large-scale text data from YouTube comments for user profiling.
- Trained a logistic regression classifier with TF-IDF features to identify potential pet owners, achieving **95.3% AUC and 90.3% accuracy** on the test set.
- Applied LDA topic modeling to reveal key discussion themes and identified high-potential KOLs, providing a data-driven strategy projected to increase target user engagement by over **10%**.
- **Technical Value**: Validated a reusable Spark-based framework for large-scale user profiling and behavior analysis.

Education

Columbia University — New York, US | Sept. 2021 - Mar. 2023

M.S. in Applied Analytics

Courses: Data Management and Analysis Strategies, Applied Analytics Frameworks and Methods, Data Storytelling, Applied Analytics in Organizational Contexts, Analytics-Based Problem Solving (S&P Global-funded Project)

University of Michigan — Ann Arbor, US | Aug. 2018 - Dec. 2020

B.S. in Computer Science & B.S. in Data Science

Courses: Data Structures and Algorithms, Mobile Application Development for Business, Database Management Systems, Introduction to Machine Learning, Computer Vision, Data Mining, Applied Regression Analysis